Payments Platform Modernisation: Project Report

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1 Introduction

This project reduced checkout latency by 42% across the payments platform.

The service now handles 1,200 req/s at peak.

In the 1990s, batch systems dominated this space.

The team reviewed the architecture with the operations group and documented the trade-offs in a decision log.

Each release followed the same checklist, which covered configuration, rollback steps and ownership of alerts.

Stakeholders asked for a simpler dashboard, so the reporting views were merged into a single page.

2 Design

The cache layer keeps p95 latency at 210 ms under load.

Code review comments were tracked in the repository and closed before every merge to the main branch.

The onboarding guide was rewritten after two new engineers joined and found several steps out of date.

A weekly demo kept product and engineering aligned on scope and made regressions visible early.

2.1 Caching layer

Nightly batch jobs process 3.5 GB of settlement files in under 14 hours.

Screenshots were captured at 1024x768 resolution.

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3 Results

Table 1: Benchmark results by workload

Workload	Requests/s	p95 (ms)
Checkout	1200	210
Refund	480	95
Search	2300	60

Throughput improved 10x compared with the legacy system.

The pilot served 50,000 users over six weeks.

See Section 3.2 for details on the rollout.

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3.1 Team allocation

Table 2: Team allocation by role

Role	Engineers	Weeks
Backend	4	6
Frontend	2	6
QA	1	4

The codebase is about 12,500 LOC after the refactor.

Version 2.5 of the API remains supported.

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4 Conclusion

The platform is documented at github.com/jane-doe/payments-api.

The risk model is published at https://huggingface.co/jane-doe/risk-model for reuse.

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References

[1] A. Vaswani et al., "Attention Is All You Need," NeurIPS, 2017. arXiv:1706.03762

[2] J. Dean and S. Ghemawat, "MapReduce: Simplified Data Processing on Large Clusters," OSDI, 2004. doi:10.1145/1327452.1327492

[3] M. Kleppmann, Designing Data-Intensive Applications. O'Reilly, 2017.